

Private World

Mission: Submit an `{ "actions": [] }` observation schedule that forms valid stereo products.

Trace:

Action: "Bash(ls case/)"

Observation: stereo files expose product geometry and pair-quality fields.

Thought: "Now I understand this problem. I need to build a stereo_imaging scheduler."

...

Action: solution =
{ "stereo_observations": [] }

Observation: "Stereo pairs: 98/141;
Coverage: 69.5%; Total conflicts: 35;
Cross target: 19"

Thought: "the evaluation framework might only care about geometric validity"

Annotation: local validation accepts what the official verifier rejects.

Observation: "Issues found: 0"

Final Result: Invalid, zero score

Calibration

Mission: Design a revisit constellation under off-nadir and slew constraints.

Trace:

...

Observation: "Diff: 17601.56 m;
Estimated score: 6.00 hours"

Thought: "Let me run the verifier to check validity"

Action: "Bash(./verifier solution.json)"

Observation: "off-nadir angle 30.041
deg exceeds 30.000 deg"

Thought: "my propagator doesn't match the verifier exactly"

Action: Uses brahe J2 numerical propagator and exact frame rotations

...

Observation: "errors: [];
num_satellites: 10;
threshold_violation_count: 0"

Final Result: Valid, high score

Adaptive Search

Mission: Add relay assets and link actions to serve all communication demands.

Trace:

Thought: "writing an empty but valid `solution.json` first"

Action: { "added_satellites": [],
"actions": [] }

Observation: "...
demand_003 76 / 90 0.844444...

demand_001 30 / 60 0.5
..."

Thought: "misses come from brief disconnected periods in the ISL graph"

...

Observation:
"unserved_demand_sample_count: 55"

Thought: "...rank candidate relays only on the 55 currently unserved samples"

...

Observation: "service_fraction = 1.0;
worst_demand_service_fraction = 1.0"

Final Result: Valid, perfect service